Information Sharing Regimes and Access to Credit

C. Fritz Foley
Harvard Business School

Agustin Hurtado
University of Maryland

Alberto Sepulveda
Financial Conduct Authority

September 21, 2026

Questions

What is the impact of public information regimes on access to credit?

Does imperfect information create incentives to lend to learn?

Does public information increase competition in credit markets?

Does this increased information limit lending to opaque borrowers?

Setting

Chile's credit card market offers a unique laboratory.

No standardized **credit score** comparable to FICO.

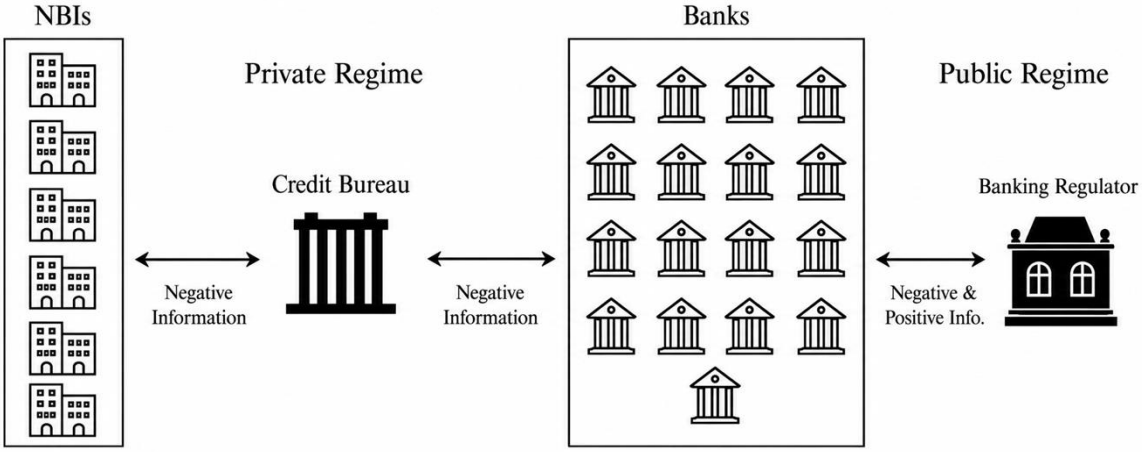
Public and private regimes operate **simultaneously**.

6 nonbank issuers (NBIs), 54% market share: Share negative information only.

17 banks, 46% market share: Also share positive information.

$\Rightarrow$ Banks cannot distinguish NBI nondefaulters from unbanked individuals.

Setting


NBIs
Private Regime
Credit Bureau
Negative Information
Negative Information
Banks
Public Regime
Banking Regulator
Negative & Positive Info.

NBI Cards ≈ Bank Cards

Both card types are open cards, accepted across merchants nationwide.

Both are subject to the same regulatory interest rate ceiling (39% in 2015).

Bankruptcy law treats loans symmetrically.

⇒ The primary distinction is the issuer type, not the card's contractual features: NBI cards function very similarly to bank cards in Chile circa 2015.

Two Regimes

Private information regimes

Relationship lending

Credit histories are **proprietary**.

Lenders earn relationship **rents**.

Rents allow them **cross-subsidize** and **learn** borrower types over time.

Public information regimes

Transactional lending

Credit histories are **shared**.

Information sharing may increase **repayment effort** and credit supply.

But it can erode **relationship rents**.

Data

Credit data construction:

On-site data collection from regulatory records.

Harmonization across NBIs and banks.

Final credit data: Panel tracking universe of borrowers across all NBIs and banks.

Two sources, linked at the individual level:

Credit records: year-month, lender ID, borrower ID, limits, balances, default.

Tax records: fiscal year, taxpayer ID, income tax bracket, gender, age.

1

Does imperfect information create incentives to lend to learn?

Hypothesized Behaviors

Nonbank issuers have an incentive to learn which potential borrowers are creditworthy

Issue cards with low limits to broad set of borrowers

Experience relatively high levels of default

Increase limits for borrowers who remain current

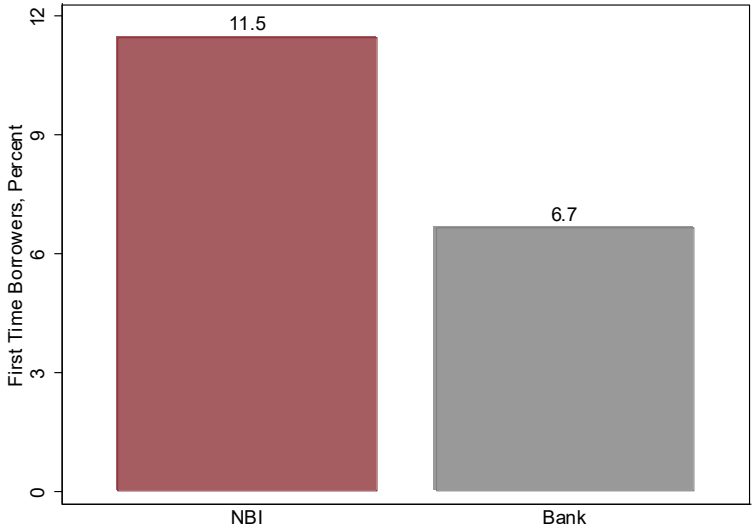
Cancel cards of borrowers who do not

Hypothesized Behaviors

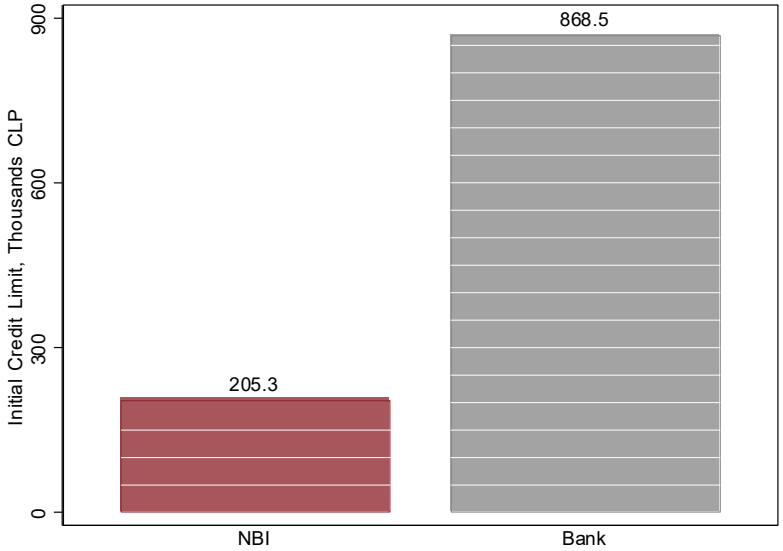
Bank issuers have no incentive to learn by lending

- Issue cards with higher limits after extensive screening
- Less likely to issue cards to potential borrowers who are difficult to evaluate

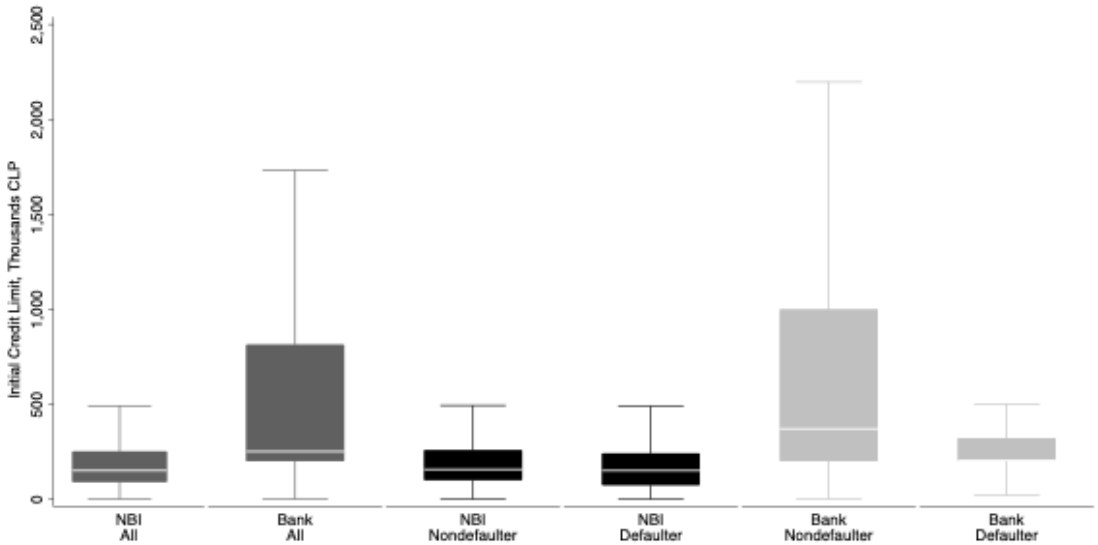
Nonbank issuers have a larger share of first-time borrowers



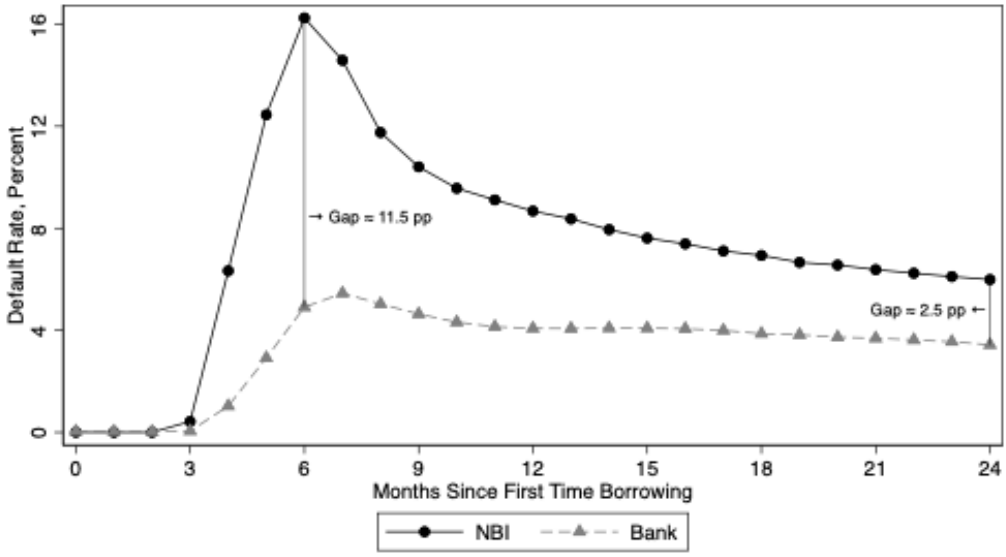
Nonbank issuers employ lower initial limits



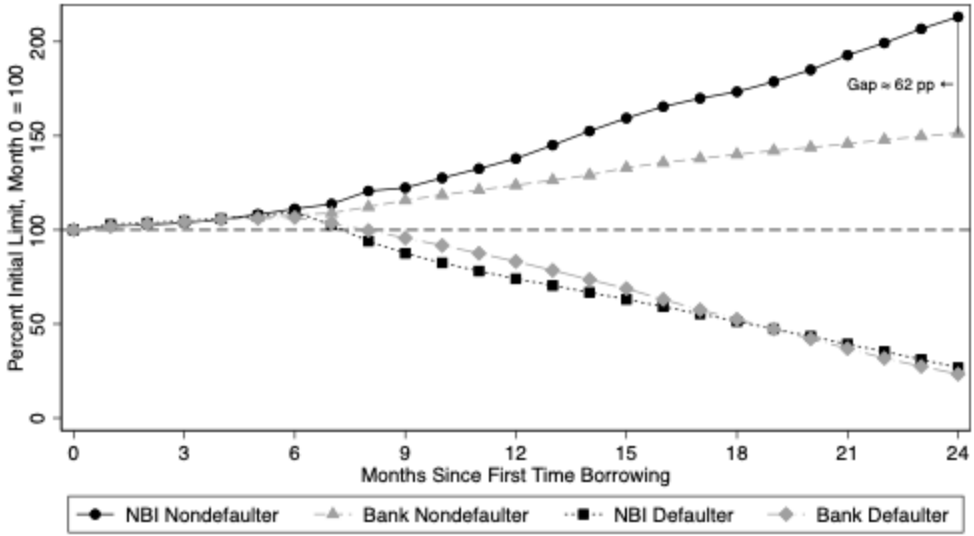
Initial limits at bank appear to reflect default risks



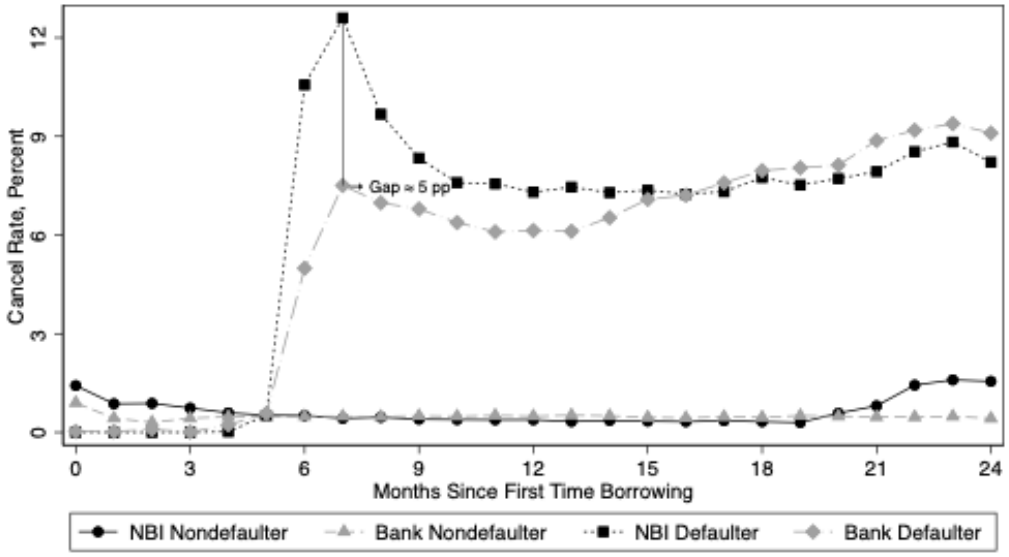
Default


Default Rate, Percent
0
4
8
12
16
→ Gap = 11.5 pp
Gap = 2.5 pp ←
0
3
6
9
12
15
18
21
24
Months Since First Time Borrowing
NBI
Bank

Credit limit growth



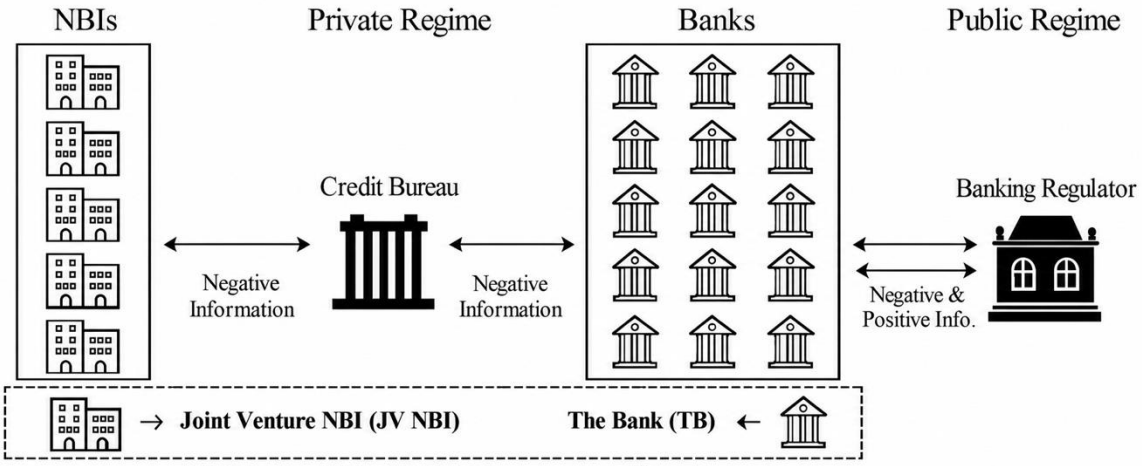
Card cancelation



2

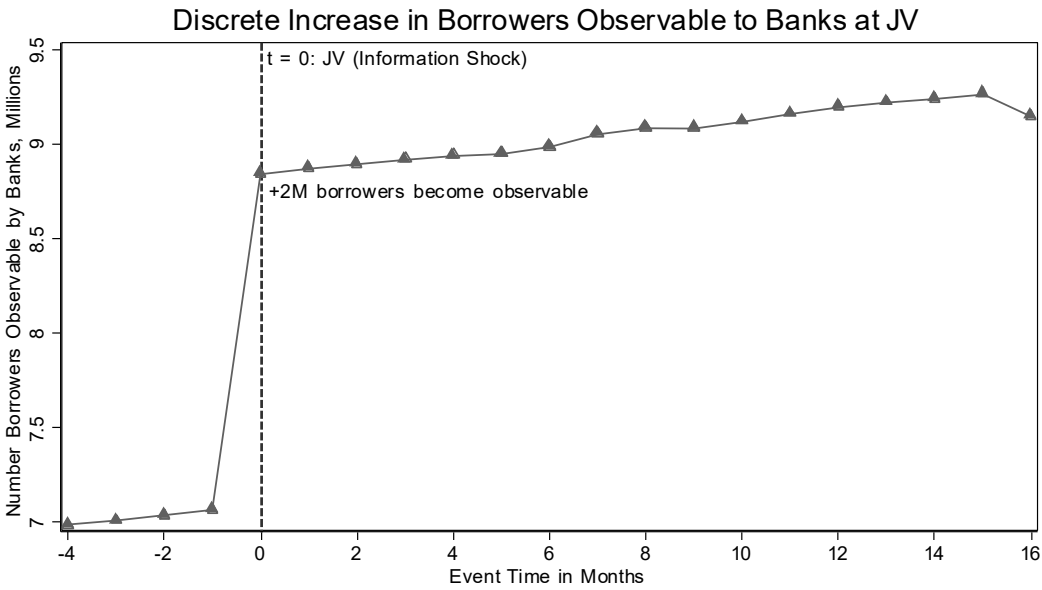
Does public information increase competition in credit markets?

Joint Venture



Joint Venture (JV): JV NBI's borrowers join public regime.

Joint Venture



Joint Venture





Impact of new information on limits of bank issued cards

Newly Observable: JV NBI borrowers, first observed by banks in May 2015.

Not Newly Observable: Other NBI borrowers, no change in the information regime.

Outcome: Total bank credit limit, excluding the JV entity.

We focus on limits because interest rates are at the regulatory cap and cannot adjust.

$$Bank\ Credit\ Limit_{i,t} = \alpha_i + \alpha_t + \sum_m \beta_m JV\ NBI_i \times 1(t = m) + \xi_{i,t}$$

Placebo: Nonbank card limits

NBIs never gain access to JV NBI borrower information.

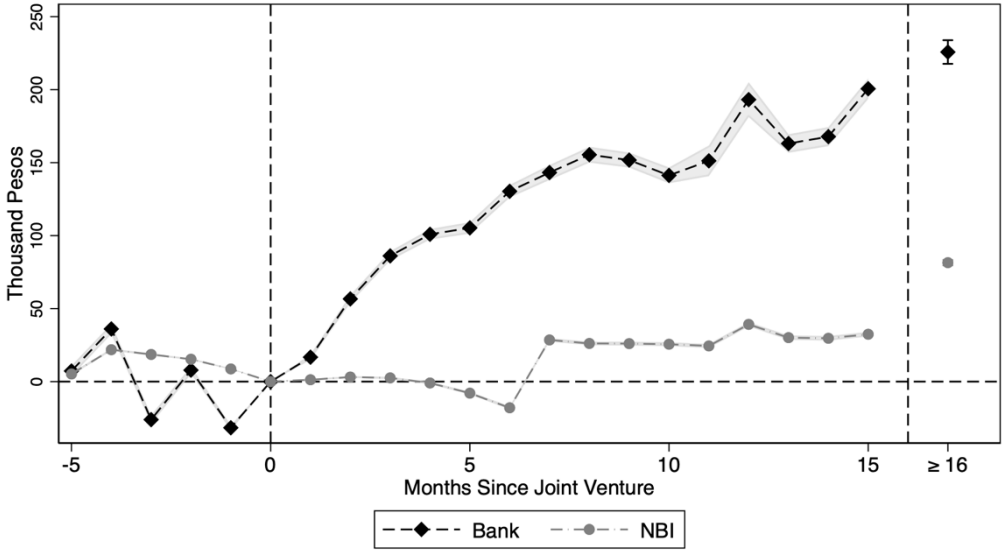
If the effect on bank limits is driven by the regime switch, NBI limits should be unaffected.

$$NBI\ Credit\ \ Limit_{i,t} = \alpha_i + \alpha_t + \sum_m \beta_m JV\ NBI_i \times 1(t = m) + \xi_{i,t}$$

Impact of new information

	Bank Credit Limits				NBI Credit Limits			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$JV\ NBI \times \mathbf{1}\{t \geq 1\}$	258.92***	180.94***			38.43***	38.25***		
	(3.09)	(3.14)			(0.67)	(0.69)		
$JV\ NBI \times \mathbf{1}\{t \in [-5,-1]\}$			-8.17***	-1.27			13.14***	13.95***
			(1.27)	(1.30)			(0.29)	(0.29)
$JV\ NBI \times \mathbf{1}\{t \in [1,5]\}$			101.49***	73.15***			-1.65***	-0.36
			(1.34)	(1.37)			(0.29)	(0.30)
$JV\ NBI \times \mathbf{1}\{t \in [6,10]\}$			198.30***	144.42***			17.82***	17.69***
			(2.39)	(2.41)			(0.58)	(0.58)
$JV\ NBI \times \mathbf{1}\{t \in [11,15]\}$			238.53***	175.20***			28.76***	31.20***
			(3.45)	(3.65)			(0.73)	(0.74)
$JV\ NBI \times \mathbf{1}\{t \geq 16\}$			320.24***	225.77***			81.63***	81.47***
			(4.04)	(4.13)			(0.93)	(0.95)
Month Fixed Effects	Y	N	Y	N	Y	N	Y	N
County × Month Fixed Effects	N	Y	N	Y	N	Y	N	Y
Outcome Mean	2,769.15	2,803.56	2,769.15	2,803.56	1,026.24	1,036.28	1,026.24	1,036.28
Observations	196,169,634	192,965,138	196,169,634	192,965,138	196,169,634	192,965,138	196,169,634	192,965,138
R-squared	0.92	0.92	0.92	0.92	0.84	0.85	0.84	0.85

Impact of new information



Is this really reflecting new information? Default prediction evidence

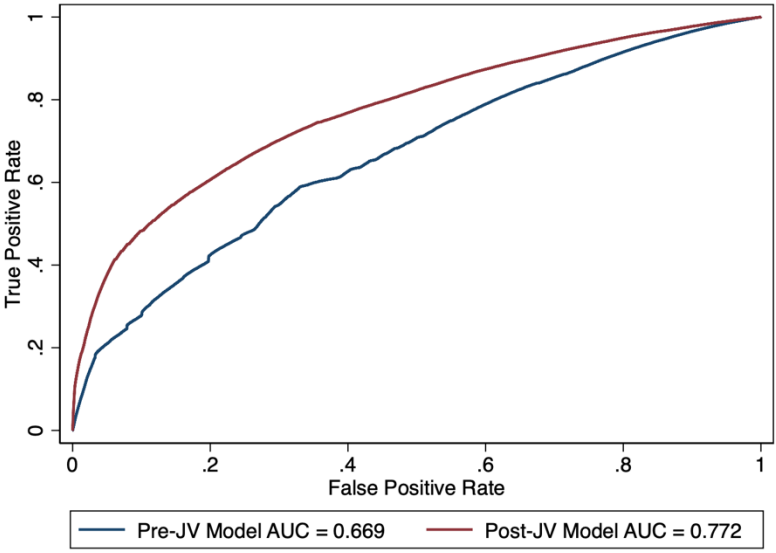
Transaction reveals credit information of JV NBI borrowers to all banks for the first time. Banks can now reassess each JV NBI borrower's creditworthiness using the new information. We approximate this reassessment by estimating pre and post default prediction models.

⇒ **Post-transaction information raises AUC from 0.67 to 0.77.**
Main drivers: JV NBI limits and balances.

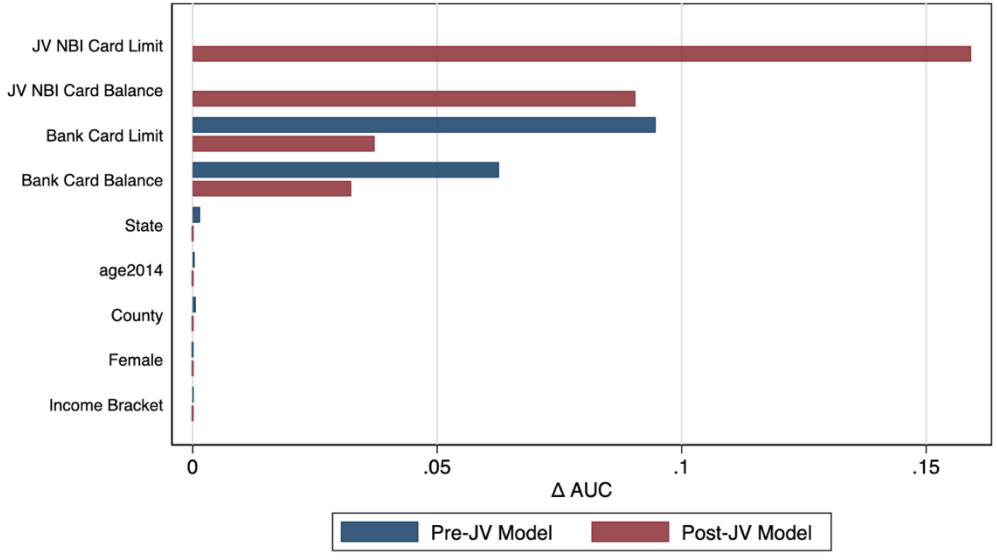
Predicting Default

	Logit Model			Probit Model		
	(1)	(2)	(3)	(4)	(5)	(6)
Bank Credit Limit	-0.67***	-0.37***	-0.38***	-0.50***	-0.28***	-0.28***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Bank Credit Balance	1.18***	0.74***	0.76***	0.99***	0.64***	0.65***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
JV NBI Credit Limit		-8.67***	-7.21***		-5.75***	-4.60***
		(0.06)	(0.06)		(0.05)	(0.04)
JV NBI Credit Balance		11.31***	7.04***		8.62***	4.53***
		(0.07)	(0.09)		(0.06)	(0.08)
JV NBI Maxed-Out Exposure			4.99***			5.89***
			(0.07)			(0.07)
Age	-1.61***	-1.16***	-1.17***	-1.58***	-1.19***	-1.18***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Female	-0.78***	-0.86***	-0.83***	-0.76***	-0.80***	-0.77***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
Income Bracket 2	-0.78***	-0.13***	-0.26***	-1.01***	-0.44***	-0.57***
	(0.05)	(0.05)	(0.05)	(0.04)	(0.05)	(0.05)
Income Bracket 3	-0.74***	0.03	-0.15*	-1.15***	-0.40***	-0.59***
	(0.08)	(0.08)	(0.08)	(0.07)	(0.08)	(0.08)
Income Bracket 4	-0.65***	0.06	-0.13	-1.04***	-0.23*	-0.47***
	(0.13)	(0.14)	(0.14)	(0.12)	(0.13)	(0.13)
Income Bracket 5	-0.13	0.45**	0.24	-0.53***	0.28	0.02
	(0.20)	(0.21)	(0.21)	(0.18)	(0.20)	(0.19)
Income Bracket 6	-0.19	0.34	0.07	-0.31	0.37	0.08
	(0.24)	(0.26)	(0.25)	(0.22)	(0.24)	(0.24)
Income Bracket 7	-0.20	0.35	0.21	-0.26	0.50	0.30
	(0.41)	(0.44)	(0.44)	(0.37)	(0.40)	(0.40)
Income Bracket 8	-0.21	0.08	-0.09	0.07	0.84*	0.47
	(0.46)	(0.47)	(0.47)	(0.42)	(0.45)	(0.44)
Default Mean	6.16	6.16	6.16	6.16	6.16	6.16
Observations	1,901,685	1,901,685	1,901,685	1,901,685	1,901,685	1,901,685
Pseudo R-squared	0.046	0.093	0.100	0.044	0.084	0.094

New information increases AUC



TN card variables drive the improvement in default prediction



Change in predicted default specification

$$Bank\ Credit\ Limit_{i,t} = \alpha_i + \alpha_t + \sum_m \beta_m \Delta PD_i \times 1(t = m) + \xi_{i,t}$$

ΔPD = predicted default estimated without JV NBI information – predicted default estimated with JV NBI information

ΔPD is larger when predicted default decreases more

i, t index individual and month-year

SEs clustered at the individual level

Change in predicted default analysis

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔPD	324.35***	319.60***			50.45***	44.19***		
	(57.31)	(58.94)			(3.10)	(3.12)		
$\Delta PD \times \mathbf{1}\{t \in [1,5]\}$			63.32**	62.36**			11.15***	9.57***
			(25.44)	(26.18)			(1.37)	(1.36)
$\Delta PD \times \mathbf{1}\{t \in [6,10]\}$			192.14***	189.90***			28.09***	24.46***
			(41.48)	(42.57)			(2.34)	(2.35)
$\Delta PD \times \mathbf{1}\{t \in [11,15]\}$			311.08***	306.73***			42.34***	36.95***
			(54.99)	(56.60)			(3.12)	(3.15)
$\Delta PD \times \mathbf{1}\{t \geq 16\}$			451.38***	444.54***			72.25***	63.44***
			(85.36)	(87.80)			(4.47)	(4.49)
Continuous ΔPD	Y	Y	Y	Y	N	N	N	N
Top Quartile ΔPD	N	N	N	N	Y	Y	Y	Y
Month Fixed Effects	Y	N	Y	N	Y	N	Y	N
County × Month Fixed Effects	N	Y	N	Y	N	Y	N	Y
Outcome Mean	126.95	128.43	126.95	128.43	126.95	128.43	126.95	128.43
Observations	13,633,800	13,425,728	13,633,800	13,425,728	13,633,800	13,425,728	13,633,800	13,425,728
R-squared	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01

3

Does this increased information limit lending to opaque borrowers?

Nonbank issuers have a larger share of young, female, low-income borrowers

	(1)	(2)	(3)
	NBI	Bank	NBI-Bank Difference
Young borrower share	10.3	9.2	1.1***
			(0.0)
Female borrower share	54.2	47.0	7.2***
			(0.1)
Low-income borrower share	92.9	71.9	21.0***
			(0.0)

Impact on composition of first-time borrowers

Sample: First-time borrowers whose lender is the JV entity or another nonbank issuer

Information Shock: The entity's information regime changes; other NBIs are unaffected.

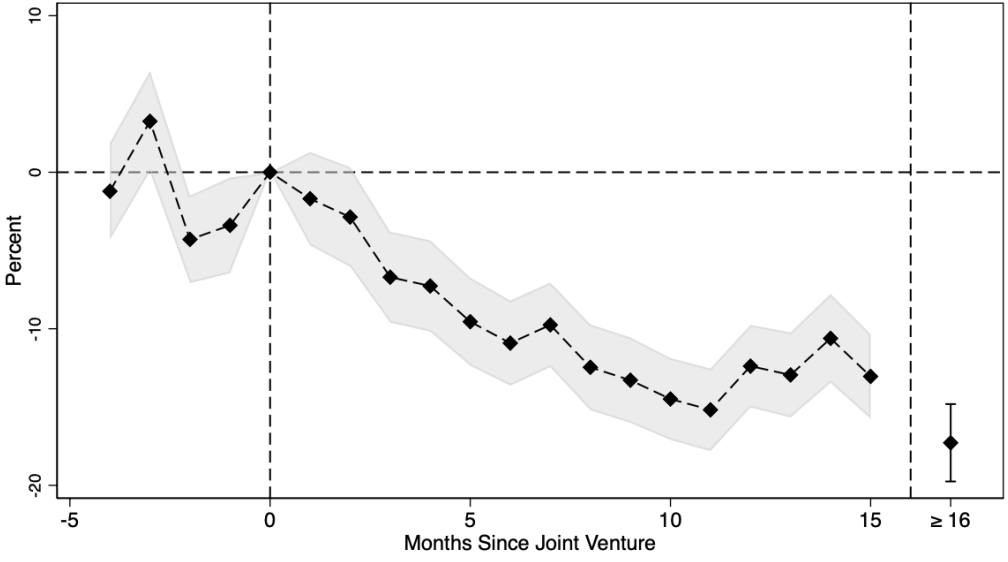
Outcome: Proxies for borrower opacity (dummies for young, female, and low-income).

$$G_{i,t} = \alpha_t + \sum_m \beta_m JV\ NBI_i \times 1(t = m) + \xi_{i,t}.$$

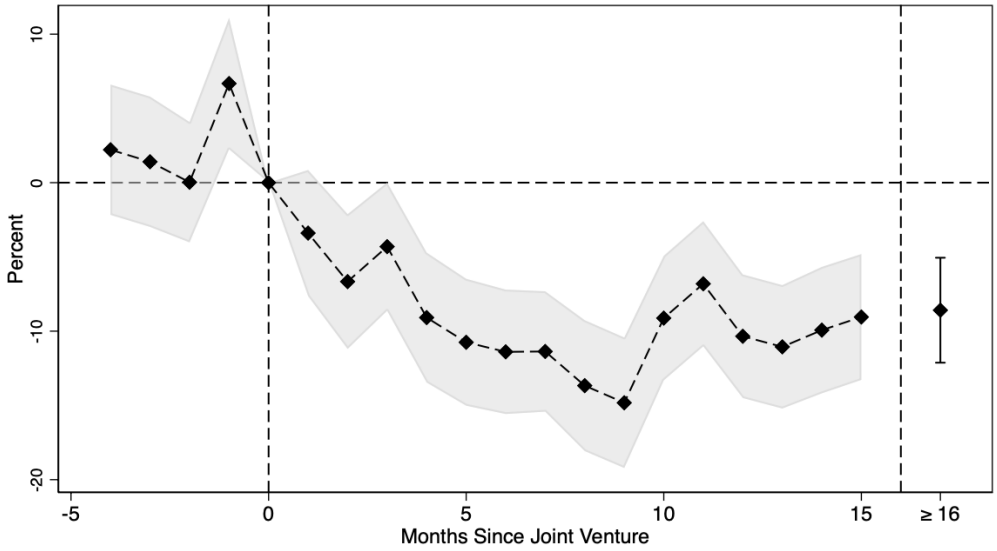
Composition of first-time borrowers

	Young Borrower				Female Borrower				Low Income Borrower			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
JV Entity × $\mathbf{1}\{t \geq 1\}$	-11.42***	-11.53***			-10.61***	-11.01***			-1.23***	-3.45***		
	(0.42)	(0.43)			(0.65)	(0.66)			(0.44)	(0.45)		
JV Entity × $\mathbf{1}\{t \in [-4,-1]\}$			-1.51	-1.72			2.27	2.13			1.59	1.47
			(1.31)	(1.32)			(1.84)	(1.87)			(1.30)	(1.32)
JV Entity × $\mathbf{1}\{t \in [1,5]\}$			-5.92***	-5.66***			-6.59***	-6.67***			-0.83	-1.02
			(1.29)	(1.30)			(1.83)	(1.85)			(1.29)	(1.32)
JV Entity × $\mathbf{1}\{t \in [6,10]\}$			-11.85***	-11.92***			-11.75***	-11.93***			-3.07**	-3.39***
			(1.26)	(1.28)			(1.81)	(1.84)			(1.29)	(1.31)
JV Entity × $\mathbf{1}\{t \in [11,15]\}$			-12.27***	-12.75***			-9.01***	-9.58***			1.51	-0.24
			(1.26)	(1.28)			(1.81)	(1.83)			(1.28)	(1.31)
JV Entity × $\mathbf{1}\{t \geq 16\}$			-14.83***	-15.33***			-8.78***	-9.87***			0.73	-3.16**
			(1.24)	(1.26)			(1.77)	(1.80)			(1.26)	(1.28)
Month Fixed Effects	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
County × Month Fixed Effects	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y
Outcome Mean	7.39	7.70	7.39	7.70	49.97	50.90	49.97	50.90	88.28	90.49	88.28	90.49
Observations	1,444,117	1,343,674	1,444,117	1,343,780	1,444,117	1,343,674	1,444,117	1,343,780	1,444,117	1,343,674	1,444,117	1,343,780
R-squared	0.01	0.03	0.00	0.01	0.00	0.02	0.00	0.01	0.01	0.02	0.01	0.01

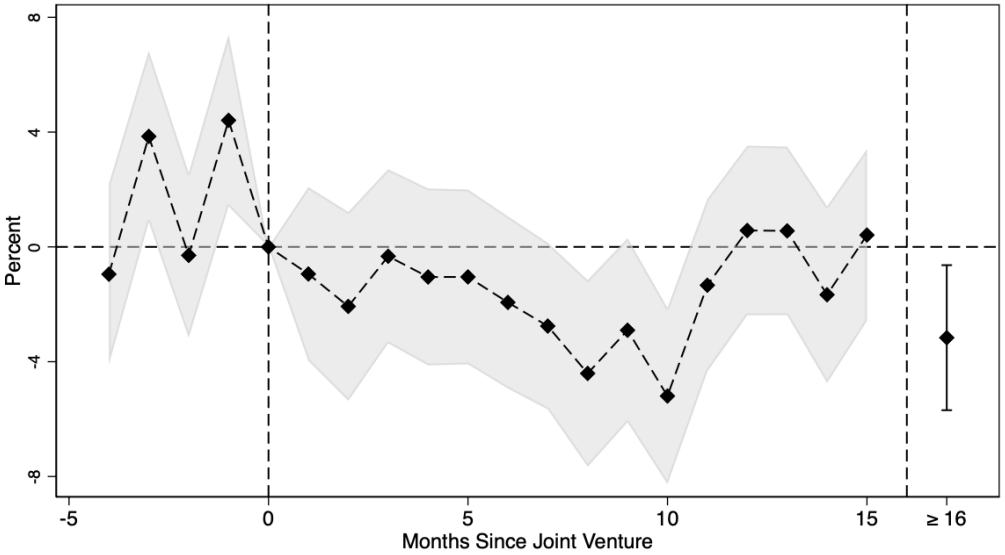
JV entity's first-time young borrowers



JV entity's first-time female borrowers



JV entity's first-time low-income borrowers



Concluding Remarks

Analysis exploits a dual regime setting and a join venture to show:

The private information regime incentivizes information production through learning by lending.

A switch to a public regime increases competition and expands access for incumbent borrowers.

But this public information regime appears to reduce incentives to originate credit to new, informationally opaque borrowers.

Thank you